Example: As a simplified illustrative example using $CTDI_{100}$, consider a multi-slice scanner. The site anticipates a workload of 100 patients per 40 h week, of which 60 patients receive one body scan and 40 patients receive one head scan. The average technique is 120 kVp, with T_b = 2 cm (*e.g.*, four 5 mm slices are acquired per rotation),

with an average scan length L for a head scan of 20 cm at 300 mAs and a pitch of unity, and L = 50 cm for a body scan at 250 mAs using a pitch of 1.35. The ${}_{n}CTDI_{100}$ values are 0.223 and 0.138 mGy mAs^{-1} for a head and body scan, respectively.

Thus, from Equation 5.6, the secondary air-kerma values per patient at 1 m for one head and one body scan, respectively, are:

$$K_{sec}^{1}\ (\text{head}) = (9 \times 10^{-5}\ \text{cm}^{-1})(20\ \text{cm})\ (300\ \text{mAs})\ (0.223\ \text{mGy mAs}^{-1})$$
$$= 0.12\ \text{mGy patient}^{-1}$$

$$K_{sec}^{1}\ (\text{body}) = (3 \times 10^{-4}\ \text{cm}^{-1}) \left(\frac{50\ \text{cm}}{1.35}\right) (250\ \text{mAs})\ (0.138\ \text{mGy mAs}^{-1})$$
$$= 0.38\ \text{mGy patient}^{-1}.$$

Therefore, for a barrier bounding an uncontrolled, fully-occupied area at 3 m distance from the isocenter, the unshielded weekly secondary air kerma for the total of 100 patients would be:

$$K_{sec} = \left(\frac{1\ \text{m}}{3\ \text{m}}\right)^2 [40\ (0.12) + 60\ (0.38)] = 3.1\ \text{mGy week}^{-1}.$$

The required barrier transmission is:

$$B = \frac{0.02\ \text{mGy week}^{-1}}{3.1\ \text{mGy week}^{-1}} = 6.5 \times 10^{-3}.$$

The required barrier thickness from the CT scanner attenuation curves (Figure A.2 in Appendix A) is:

$$x_{\text{barrier}} = 1.3\ \text{mm lead}.$$

Thus a 1/16 inch standard lead thickness (1.59 mm) is adequate for this barrier under the conditions stated above. It is important to note that the shielding provided by the standard concrete floor and ceiling in a modern multi-slice scanner room may not be adequate (Langer and Gray, 1998).

5.6.1 *Dose-Length Product Method*

Computed tomography is currently undergoing rapid and significant change. Several CT manufacturers now display *DLP*